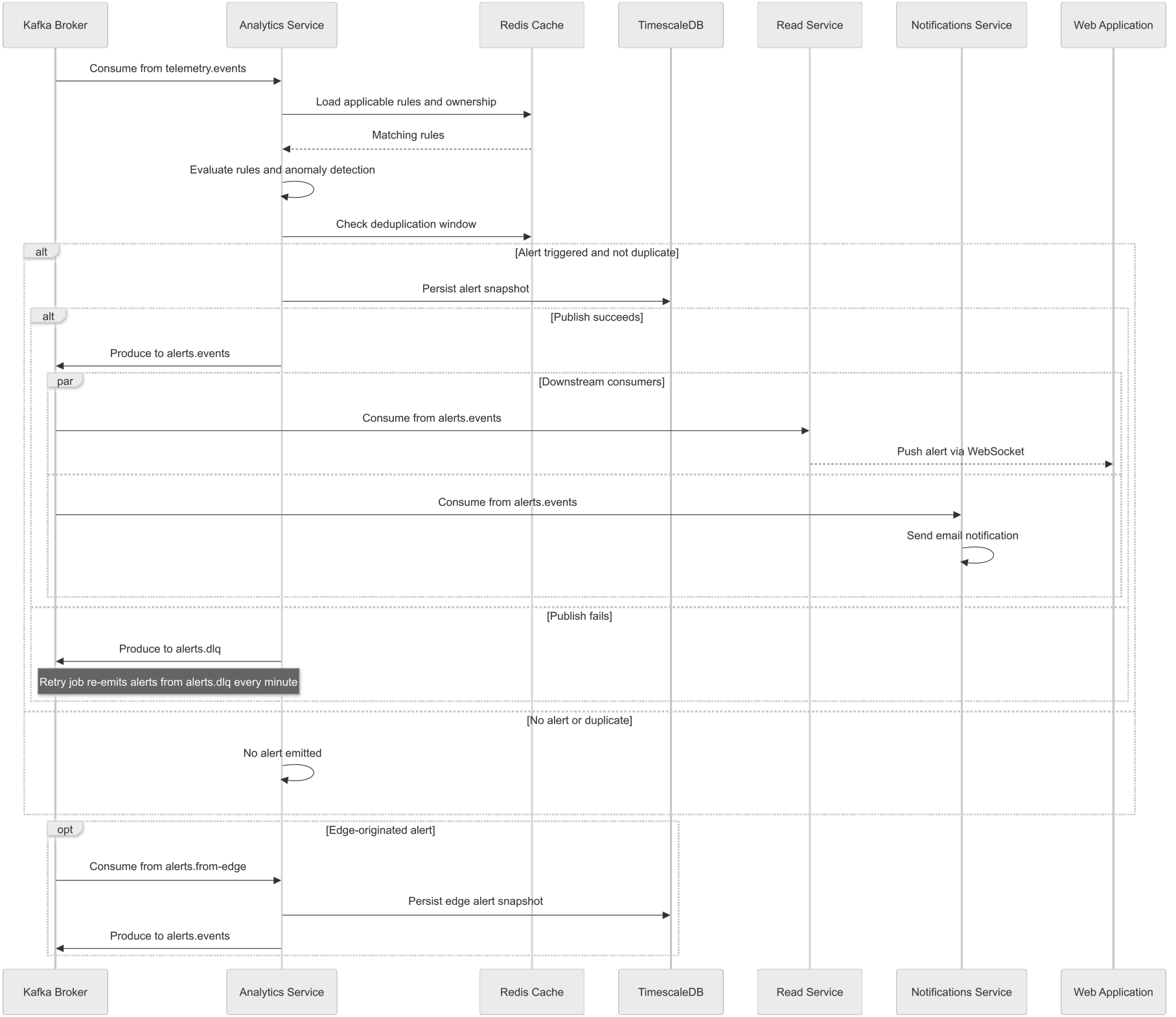
Kafka Broker
Analytics Service
Redis Cache
TimescaleDB
Read Service
Notifications Service
Web Application
Consume from telemetry.events
Load applicable rules and ownership
Matching rules
Evaluate rules and anomaly detection
Check deduplication window
alt
[Alert triggered and not duplicate]
Persist alert snapshot
alt
[Publish succeeds]
Produce to alerts.events
par
[Downstream consumers]
Consume from alerts.events
Push alert via WebSocket
Consume from alerts.events
Send email notification
[Publish fails]
Produce to alerts.dlq
Retry job re-emits alerts from alerts.dlq every minute
[No alert or duplicate]
No alert emitted
opt
[Edge-originated alert]
Consume from alerts.from-edge
Persist edge alert snapshot
Produce to alerts.events
Kafka Broker
Analytics Service
Redis Cache
TimescaleDB
Read Service
Notifications Service
Web Application